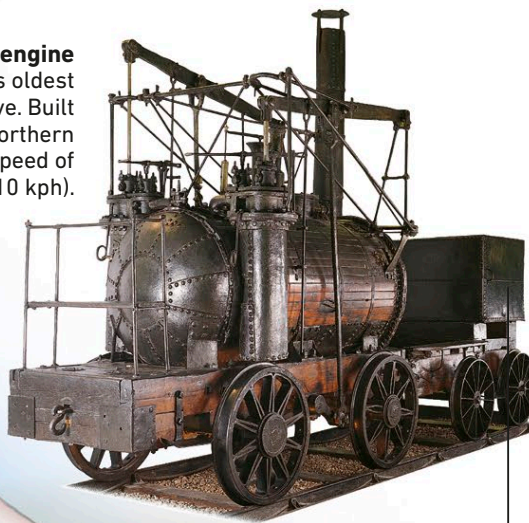


Early steam engine

Puffing Billy is the world's oldest surviving steam locomotive. Built in 1813 to haul coal in northern England, it had a top speed of about 6 mph (10 kph).



Coal carried in the tender was burned to heat water in the boiler and produce steam to drive the wheels.



Bullet-shaped nose enables locomotive to cut through the air more easily, increasing speed.

Infrastructure

The built and engineered systems that we rely on every day—from sewers and telecommunication networks to power lines, railroads, and roads—are collectively known as infrastructure. Without such systems, our modern way of life would be impossible.

- **First telephone exchange**

The first commercial exchange to connect callers was built in New Haven, Connecticut, in 1878.

- **Intercity railroad**

Opened in 1830, the Manchester to Liverpool route in England was the first intercity railroad.

**Ulm–Stuttgart autobahn, 1950**

Germany was a pioneer of the freeway, or autobahn, in the 1930s. Cars did not clog the roads until much later!

Construction

A steel-and-concrete building revolution began in the late 19th century. Frames made of steel girders allowed taller structures to be built, and the invention of reinforced concrete—concrete with steel rods set into it—introduced an amazingly strong, durable new material. Together, steel and reinforced concrete gave birth to the modern skyscraper, changing the face of the world's cities.

- **Ancient concrete**

The Romans were experts in building with concrete. It was used in the construction of the Colosseum and the Pantheon in Rome.

- **World's oldest skyscraper city?**

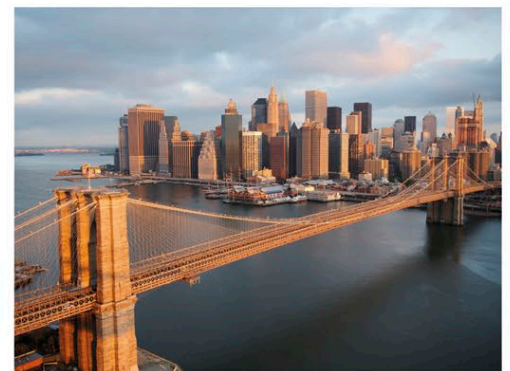
Shibam, in Yemen, has about 500 high-rise apartment buildings made of mud brick, most dating from the 16th century.

- **First steel-framed skyscraper**

Completed in 1885, the innovative 10-story Home Insurance Building in Chicago, Illinois, used a steel frame to support the walls.

- **Reinforced first**

The first skyscraper built with reinforced concrete was the 15-story Ingalls Building, in Cincinnati, Ohio, erected in 1903.

**Manhattan, then and now**

The Brooklyn Bridge spans New York's East River. The view across to Manhattan Island has changed dramatically since the bridge opened in 1883, and it now bristles with skyscrapers.